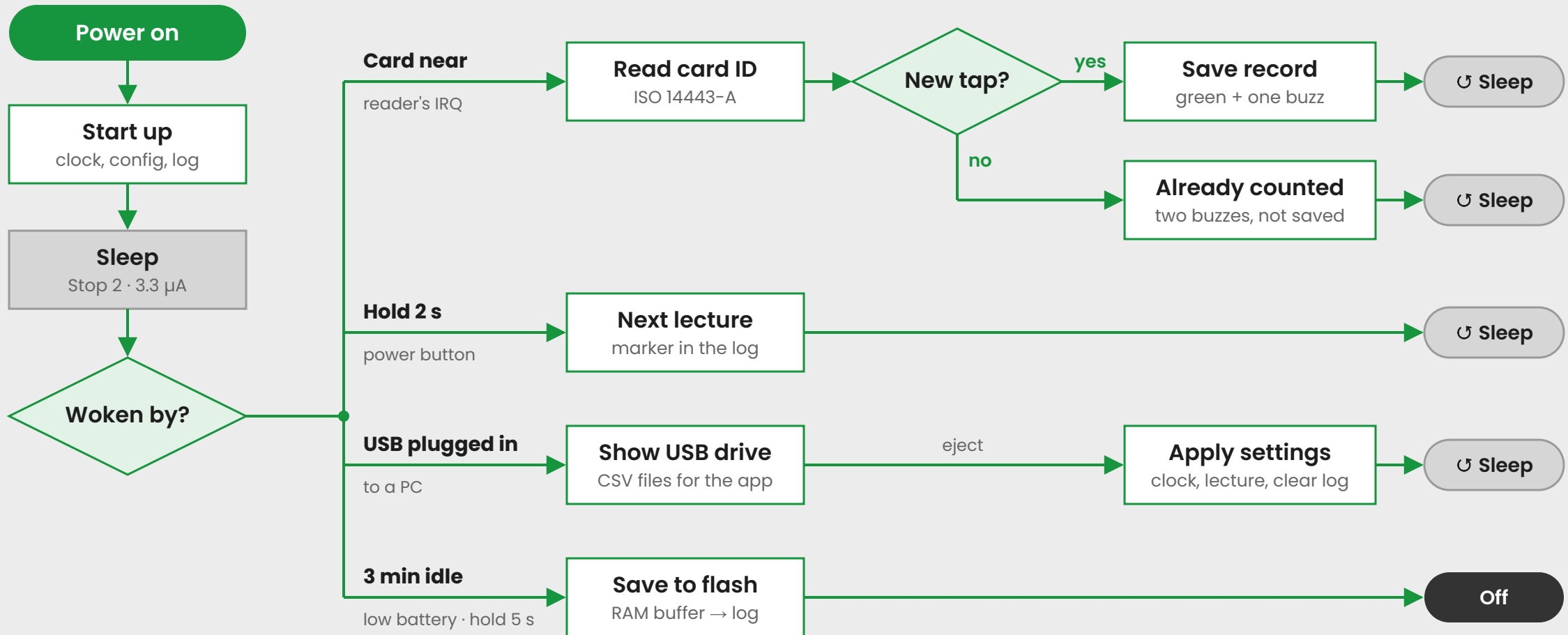


Firmware Flow

The MCU sleeps in **Stop 2** until something wakes it, handles it, and goes back to sleep.



Interrupt priority

5 USB

6 card IRQ · button · VBUS · wake timer

15 SysTick

lower number wins; USB answers the PC, the rest only wake the main loop

New tap = not already in this lecture and not the same card within 10 s. Records wait in RAM and go to flash 5 s after the last tap. The reader looks for cards by itself while the MCU sleeps.